

DLC

Daily Leverage Certificates

Country of Emission: Singapore

Executive Summary:

This document serves as a detailed technical companion to the primary educational video on Daily Leverage Certificates (DLCs) listed on the Singapore Exchange (SGX). First introduced on SGX in July 2017 making Singapore the first ASEAN exchange to offer them. DLCs are structured leveraged products issued primarily by Société Générale, offering fixed leverage factors of 2x, 3x, 5x, or 7x on underlying assets such as the Straits Times Index (STI), Hang Seng Index (HSI), S&P 500, and select Singapore-listed equities.

This project is structured as three complementary deliverables: a video introduction accessible to a broad audience, this written technical guide for readers seeking depth, and an interactive model enabling hands-on exploration of DLC pricing dynamics. While the video provides an accessible walkthrough of the product its purpose, mechanics, and investor suitability, this written guide is intended for readers who require a more granular, step-by-step understanding of how DLCs function in practice. Alongside this written guide, an interactive pricing model is provided, allowing readers to input their own parameters, leverage factor, underlying price movements, and holding period and observe in real time how DLC value, compounding, and airbag triggers respond.

The following pages dissect the daily reset mechanism, illustrate the impact of compounding and path dependency through worked numerical examples, and outline the embedded risk controls, namely the airbag mechanism and mandatory stop-loss features. The document also covers the regulatory framework governing DLCs under the Monetary Authority of Singapore (MAS), including the Specified Investment Products (SIP) classification. A comparison with similar leveraged instruments such as CFDs, structured warrants, and leveraged ETFs is also provided.

By pairing a concise video narrative with this comprehensive written document, we aim to address the needs of a diverse audience: clients seeking to understand a product, finance students bridging theory with real-world instruments, or internal stakeholders evaluating structured product design.

Table of Contents

1	INTRODUCTION TO DAILY LEVERAGE CERTIFICATES (DLC)	3
1.1	KEY CHARACTERISTICS (LEVERAGE FACTORS, BOARD LOT, NO MARGIN CALLS)	3
1.2	LONG VS SHORT DLCs	3
2	HOW TO READ A DLC NAME	4
3	MECHANICS: THE DAILY RESET	4
3.1	HOW THE DAILY RESET WORKS	4
3.2	INTRINSIC CLOSE VS LAST TRADED PRICE	5
3.3	THE MATHEMATICS OF THE DAILY RESET	5
4	THE COMPOUNDING EFFECT	6
4.1	COMPOUNDING IN TRENDING MARKETS (POSITIVE CASE)	6
4.2	COMPOUNDING IN VOLATILE / SIDEWAYS MARKETS (NEGATIVE CASE)	6
5	THE AIRBAG MECHANISM	7
5.1	PURPOSE AND TRIGGER LEVELS BY LEVERAGE FACTOR	7
5.2	LIMITATIONS OF THE AIRBAG	9
6	COSTS AND FEES	10
6.1	INTRADAY COSTS	10
6.2	OVERNIGHT COSTS	11
7	REGULATORY FRAMEWORK (SINGAPORE)	12
7.1	MAS OVERSIGHT AND SIP CLASSIFICATION	12
7.2	CUSTOMER KNOWLEDGE ASSESSMENT (CKA) / CUSTOMER ACCOUNT REVIEW (CAR)	12
7.3	SGX TRADING HOURS AND MARKET MAKER OBLIGATIONS	12
8	COMPARISON WITH SIMILAR PRODUCTS	13
8.1	DLC vs CFD	13
8.2	DLC vs STRUCTURED WARRANTS	13
8.3	DLC vs LEVERAGED ETFs	14
9	HOW THE ISSUER CREATES AND HEDGES DLCS	14
9.1	CONSTRUCTING THE DLC: THE DELTA HEDGE	14
9.2	DAILY REBALANCING OF THE HEDGE	15
9.3	MANAGING OVERNIGHT GAP RISK	15
9.4	THE AIRBAG AS AN ISSUER RISK CONTROL	16
10	DLC TRADING STRATEGIES	16
10.1	TACTICAL DIRECTIONAL TRADE	16
10.2	PORTFOLIO HEDGING WITH A SHORT DLC	17
11	CONCLUSION	18
12	GLOSSARY	18
13	REFERENCES	20

1 Introduction to Daily Leverage Certificates (DLC)

Daily Leverage Certificates (DLCs) are exchange-traded financial products that enable investors to take a leveraged exposure to an Underlying Asset, such as an equity index or a single stock. First introduced on the Singapore Exchange (SGX) in July 2017, DLCs made Singapore the first ASEAN exchange to offer this product. Because DLCs are listed on the cash market of the SGX, they trade like regular stocks, no special account type is required beyond the knowledge of the product. The primary issuer of DLCs on SGX is Société Générale, with Macquarie also active as an issuer.

1.1 Key Characteristics (Leverage factors, board lot, no margin calls)

Daily Leverage Certificates replicate the performance of an Underlying Asset versus its previous day closing level, with a fixed leverage factor. The leverage effect means that any movements in the Underlying Asset are amplified. DLCs offer fixed leverage factors of 2x, 3x, 5x or 7x. Therefore, the daily percentage return of a DLC will be equal to the daily percentage return of the Underlying Asset multiplied by the fixed leverage factor, before costs and fees:

$$\text{DLC Daily Performance (in \%)} = (\text{Underlying Asset Daily Performance (in \%)} \times \text{Leverage Factor}) - \text{Costs \& Fees}$$

Each DLC is a separate trading counter on the exchange, with a separate counter name, trading code and price. The price of a DLC can be different from its Underlying Asset and is decided by the Issuer on the initial issue date. After a DLC is listed, its performance will follow the pre-determined relationship as described above.

The available underlying assets on SGX include major indices such as the Straits Times Index (STI), Hang Seng Index (HSI), Hang Seng Tech Index (HSTECH), S&P 500, and Nasdaq-100, as well as select Singapore-listed single stocks including DBS, OCBC, UOB, and Singtel. DLCs can be traded on a per board lot basis, equivalent to 100 certificates. They have no margin maintenance requirements, meaning less capital is locked up and investors retain more flexibility in their asset allocation. The maximum loss is limited to the initial invested capital. DLCs have a maximum tenure of three years, after which they mature and are delisted.

1.2 Long vs Short DLCs

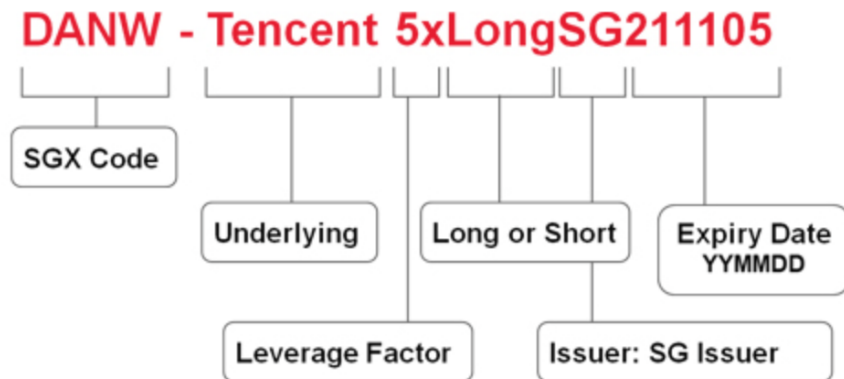
There are two types of Daily Leverage Certificates:

- Daily Long: profits when the Underlying Asset rises in value
- Daily Short: profits when the Underlying Asset falls in value

They enable investors to take a long or short leveraged exposure to an Underlying Asset without the need for a margin account or short-selling permissions. This makes them particularly accessible compared to futures or CFDs, where margin requirements and the risk of margin calls apply. If a DLC is held until expiry, it will be settled at fair value based on the pricing formula, which applies the leverage factor to the cumulative daily performance of the Underlying Asset from the Intrinsic Close of each preceding Trading Day, net of costs and fees.

2 How to Read a DLC Name

Each DLC listed on SGX follows a standardized naming convention that encodes key product information directly into its name. Understanding this structure allows investors to immediately identify the core characteristics of a DLC at a glance, without needing to consult the issuer's documentation.



Reading the full name together: this is a Société Générale-issued DLC, traded under SGX code DANW, offering 5x leveraged long exposure to Tencent, expiring on 5 November 2021.

3 Mechanics: The Daily Reset

The daily reset is the defining structural feature of DLCs and the mechanism from which all other properties compounding, path dependency, and the airbag directly follow. Understanding it precisely is essential before engaging with any other aspect of the product.

3.1 How the daily reset works

At the end of each Trading Day, the exposure of a DLC is reset back to its fixed leverage factor relative to the closing level of the Underlying Asset. When markets open the next day, both the Underlying Asset and the DLC are measured from that closing level, not from the original purchase price of the investor.

In practical terms, this means the DLC behaves as if it is a fresh position each morning. Each day's performance is calculated independently from the prior day's close, and any gain or loss from that day is locked in before the next day's calculation begins. This is fundamentally different from a static leveraged position. Consider a hypothetical leveraged loan that gives you 5x exposure to an asset: if the asset falls 10% on Day 1, your 5x position loses 50%, but the next day's gains or losses are still calculated on your original notional amount. With a DLC, the notional resets, so Day 2's performance is 5x the move from the new, lower closing level. This daily rebalancing is what creates the compounding effect explored in Section 4.

3.2 Intrinsic Close vs last traded price

When calculating daily performance, investors must not use the DLC's last traded price from the previous session as their reference point. This is a common source of confusion and can lead to significant misreading of returns. The reason is straightforward: the last trade executed in a session may have occurred several minutes before the official market close of the Underlying Asset. If the Underlying Asset continued to move between that last trade and the official close, the last traded DLC price will not reflect that final movement.

Instead, the correct reference point is the Intrinsic Close, the theoretical fair value of the DLC calculated by the issuer at the official close of the Underlying Asset's market, applying the leverage factor to the closing level of the Underlying Asset net of daily costs. It is the Intrinsic Close, not the last traded price, that forms the base for the next day's performance calculation. As an example, if the HSI closes at 18,000 and a 5x Long DLC has an Intrinsic Close of SGD 1.20, then on the following day a 1% rise in HSI from 18,000 would produce approximately a 5% gain in the DLC calculated from SGD 1.20, not from whatever price the DLC last traded at near the end of the previous session.

3.3 The Mathematics of the Daily Reset

The intuitive explanation in Sections 3.1 and 3.2 establishes that the DLC resets its leverage each day from the Intrinsic Close. This section formalizes that mechanic mathematically, providing the foundation for understanding why compounding causes DLC returns to diverge from a simple multiple of the underlying.

$$V_n = V_0 \times \prod_{t=1}^n (1 + L \cdot R_t - C_t)$$

Where V_0 is the initial DLC value (Intrinsic Close on the day of purchase) and $\prod_{t=1}^n$ denotes the product of n successive daily return factors.

- L is the leverage factor (2, 3, 5, or 7),
- R_t the percentage return of the Underlying Asset on day t, and
- C_t the total daily cost rate on day t, representing the sum of all applicable fees.

This is fundamentally different from what an investor might naively expect. If the underlying asset moves a total of R_{total} over n days, a simple leveraged instrument would return:

$$V_n^{naive} = V_0 \times (1 + L \cdot R_{total})$$

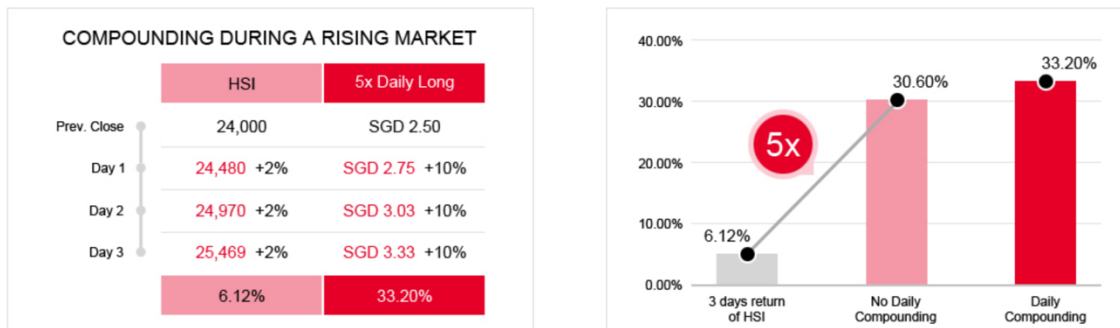
The gap between V_n and V_n^{naive} is the compounding divergence, it can be positive (trending markets) or negative (volatile markets), and its magnitude grows with both the leverage factor and the holding period.

4 The Compounding Effect

As established in Section 3, the DLC's exposure is reset to its fixed leverage factor at the end of every Trading Day. A direct consequence of this daily reset is compounding, the phenomenon where each day's return is applied not to the original investment value, but to the accumulated value from all prior days. Over multiple days, this causes the DLC's total return to diverge from a simple multiple of the underlying asset's total return. Depending on market conditions, this divergence can work in the investor's favor or sharply against it.

4.1 Compounding in trending markets (positive case)

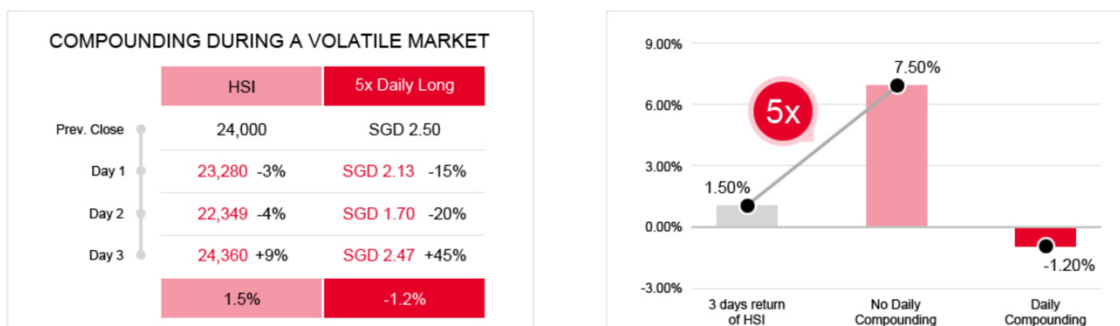
When an underlying asset moves consistently in one direction over multiple days, compounding amplifies the DLC's total return beyond a simple multiple of the underlying's total return. Consider a 5x Long DLC on the Hang Seng Index (HSI), starting at a previous close of 24,000 with the DLC priced at SGD 2.50. The HSI rises by approximately 2% each day for three consecutive days:



Each day, the 10% DLC return is applied to the previous day's accumulated DLC value, not to the original SGD 2.50. By Day 2, the 10% gain is applied to SGD 2.75, and by Day 3 to SGD 3.03. This is why the DLC outperforms even a naive 5× calculation in a steadily rising market.

4.2 Compounding in volatile / sideways markets (negative case)

The same mechanism that benefits investors in trending markets becomes a structural drag in volatile or sideways markets. When an underlying asset oscillates, falling then partially recovering, the DLC suffers a cumulative loss even if the underlying ends slightly positive. This effect is known as volatility decay or beta slippage. Consider the same 5x Long DLC, starting again at HSI 24,000 and DLC price SGD 2.50. The HSI falls 3% on Day 1, falls a further 4% on Day 2, then recovers 9% on Day 3:



The HSI ends the three-day period slightly positive at +1.50%, yet the DLC investor has lost 1.20% of their investment. The reason is mathematical asymmetry: on Day 1, the 15% loss reduces the DLC from SGD 2.50 to SGD 2.13. The subsequent 45% gain on Day 3 is then applied to this much smaller base of SGD 1.70 and is simply not large enough to recover the ground lost over the first two days.

This is the critical risk of holding DLCs over multiple days in volatile or range-bound markets. DLCs are structurally designed for short-term, directional trading not as buy-and-hold instruments. The longer the holding period in a choppy market, the more pronounced the decay.

5 The Airbag Mechanism

The Airbag Mechanism is a risk control feature built directly into every DLC. Its purpose is to reduce the DLC's effective exposure to the Underlying Asset in the event of a significant adverse intraday move limiting, though not eliminating, the risk of total capital loss within a single Trading Day. It is one of the key structural differences between DLCs and other leveraged instruments such as CFDs or leveraged futures.

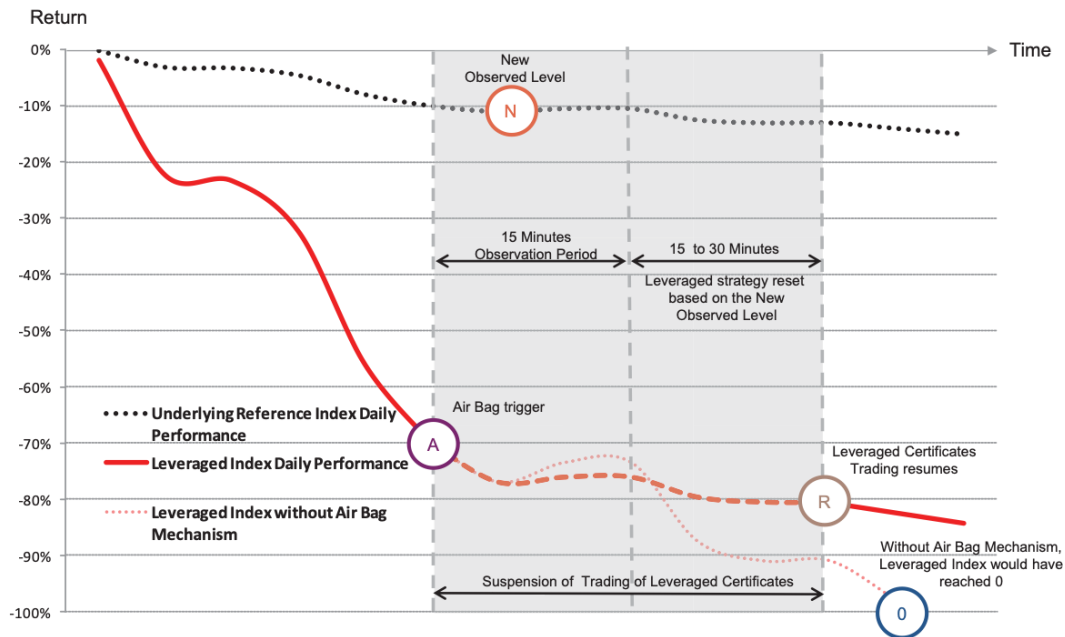
5.1 Purpose and trigger levels by leverage factor

The Airbag is triggered when the Underlying Asset moves against the DLC's direction by a pre-set threshold, measured from the previous day's closing level or from the most recent New Observed Level if the Airbag has already been triggered earlier that same day. The trigger thresholds vary depending on both the leverage factor and the type of Underlying Asset:

Underlying Type	Product	Airbag Trigger
Equity Index	2x Daily Long	- 20%
Equity Index	2x Daily Short	+ 20%
Equity Index	3x Daily Long	- 20%
Equity Index	3x Daily Short	+ 20%
Equity Index	5x Daily Long	- 10%
Equity Index	5x Daily Short	+ 10%
Equity Index	7x Daily Long	- 10%
Equity Index	7x Daily Short	+ 10%
Single Stock	3x Daily Long	- 20%
Single Stock	3x Daily Short	+ 20%
Single Stock	5x Daily Long	- 15%
Single Stock	5x Daily Short	+ 15%

One important observation from this table. Higher leverage factors have lower trigger thresholds, a 5x DLC on an equity index triggers at -10%, whereas a 3x DLC on the same index triggers at -20%. This reflects the fact that higher leverage amplifies losses faster, making earlier intervention necessary. Additionally, the Airbag can only be triggered when the underlying market is open. For example, a DLC on the HSI can only trigger its Airbag during Hong Kong Exchange (HKEx) trading hours, and a DLC on the S&P 500 only during US market hours, even if the DLC itself is still trading on SGX at that time.

The following diagram illustrates the full sequence of events when the Airbag is triggered on a Long DLC:



The mechanism unfolds in four stages:

- **Stage 1: Normal trading**

The DLC trades normally, with its price tracking 5× the daily performance of the Underlying Asset from the previous close. Both the underlying (dotted line) and the DLC (solid red line) decline steadily.

- **Stage 2: Airbag trigger (point A)**

The Underlying Asset reaches the pre-set trigger level, a -10% move for a 5x equity index DLC. At this point, trading of the DLC is immediately suspended on SGX. Without the Airbag, the DLC would continue falling toward zero (shown by the dashed red line).

- **Stage 3: 15-minute observation period**

While the DLC is suspended, the underlying market continues to trade. During this 15-minute window, the lowest level reached by the Underlying Asset is recorded, this is called the **New Observed Level (N)**. For a Short DLC, the highest level would be recorded instead. This New Observed Level becomes the new base from which the DLC performance will be calculated, resetting the leverage exposure to a lower notional.

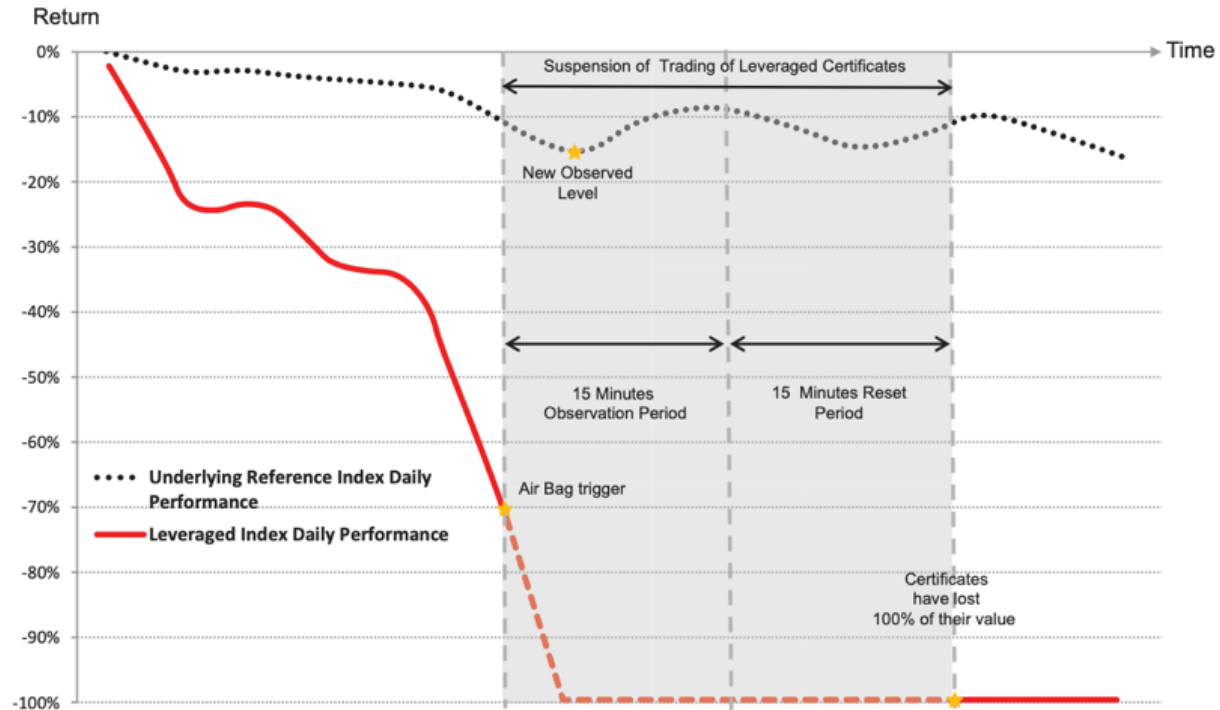
- **Stage 4: Reset and resumption (point R)**

Between 15 and 30 minutes after the Airbag triggered, the DLC strategy is reset based on the New Observed Level and trading resumes on SGX. From this point forward, the leverage factor is applied to moves in the Underlying Asset measured from the New Observed Level, not from the original previous close. Because the notional has been reduced, adverse moves have a proportionally smaller impact on the DLC's value.

The Airbag can be triggered more than once on the same Trading Day. If it triggers again, the process repeats from the most recent New Observed Level.

5.2 Limitations of the airbag

While the Airbag meaningfully reduces the risk of intraday total loss, it has important limitations that investors must understand.



It cannot protect against overnight gaps. The Airbag only activates when the underlying market is open. If the Underlying Asset gaps down sharply at the open, for example, opening 20% lower than the prior close due to overnight news the Airbag cannot intervene before the DLC is already deeply impaired. For a 5x Long DLC on an equity index, an overnight gap of 20% or more can wipe out the DLC's entire value before the market even opens.

It also cannot protect against a sharp move during the observation period itself. If the Underlying Asset continues to fall severely during the 15-minute observation window by the same threshold or more, the DLC can still lose all its value despite the Airbag having triggered. The diagram above illustrates this scenario.

Finally, it can work against investors when the market quickly rebounds. If the Underlying Asset triggers the Airbag and then rapidly recovers, the DLC misses part of that recovery. The rebound gains are applied from the New Observed Level, not from the level the market has already bounced back to. This means the Airbag can lock in losses that the market itself has erased by the time trading resumes.

In summary, the Airbag reduces but does not eliminate the risk of total capital loss. It is a partial protection mechanism, not a guarantee.

However, beyond the Airbag, DLCs carry several additional structural features investors should be aware of:

- **Limited lifespan:** DLCs have a maximum tenure of three years from the date of listing, after which they mature and are settled at fair value
- **Delisting:** A DLC will be delisted before maturity if it loses all its value due to extreme market movements, typically when the leveraged loss exhausts the entire invested capital. Delisting can also occur at the issuer's discretion in certain circumstances.
- **Adjustments for corporate actions:** If the Underlying Asset of a DLC is a single stock and that company undertakes a corporate action, such as a bonus issue, rights issue, or stock split, the DLC issuer will assess whether the action has a dilutive or concentrative effect on the share value. An adjustment will then be made to preserve the economic equivalent value of the DLC, with an effective date determined by the issuer.

6 Costs and Fees

Investing in DLCs involves several layers of costs and fees, which differ depending on whether the position is closed within the same Trading Day or held overnight. Understanding the fee structure is essential since these costs reduce the DLC's daily return and compound over time in the same way that returns do. The table below summarizes which fees apply in each scenario:

	Trading Intra-Day	Held Overnight
Brokerage Fee	Yes	Yes
Bid / Ask Spread	Yes	Yes
Funding Cost	No	Yes
Stock Borrowing Costs (Short DLC only)	No	Yes
Rebalancing Costs	No	Yes
Gap Premium	No	Yes
Management Fee	No	Yes

6.1 Intraday costs

Investors who open and close their DLC position within the same Trading Day, without holding overnight incur only two categories of cost:

- **Brokerage Fee:** A commission charged by the broker for executing the trade, applied both on entry and on exit. This is structurally identical to trading equities on SGX and is charged on the value of the DLC position, not on the total leveraged exposure. This is an important distinction from CFDs, where commission is typically charged on the full notional.
- **Bid/Ask Spread:** The difference between the price at which the market maker is willing to buy and the price at which they are willing to sell the DLC at any given moment. This is an implicit cost paid on both entry and exit, and its size reflects the liquidity of the DLC. Market makers on SGX are obligated to provide continuous bid and ask quotes during trading hours, which keeps spreads competitive under normal conditions.

6.2 Overnight costs

Investors who hold a DLC position beyond the close of a Trading Day incur the full range of fees. These are deducted from the DLC's value at the close of each Trading Day and fall into two categories: **Leverage Strategy Fees** and **Hedging Fees**.

Leverage Strategy Fees are costs associated with maintaining the leveraged exposure:

- **Funding Cost:** the cost of financing the leveraged position overnight. This is benchmarked to a short-term interest rate. For SGX-listed DLCs on Singapore underlyings, this would typically reference SORA or an equivalent Singapore rate.
- **Rebalancing Cost:** a cost of approximately 0.10% per annum, reflecting the operational cost of rebalancing the hedge at the end of each Trading Day to restore the leverage factor to its fixed level. This cost is a function of both the leverage factor and the daily performance of the underlying.
- **Stock Borrowing Cost (Short DLCs only):** for Short DLCs, the issuer must borrow the underlying stock or index components to construct the short hedge. The cost of this borrowing is passed through to the investor and varies by underlying.

Hedging Fees are costs associated with the overnight risk protection built into the DLC:

- **Gap Premium:** a daily deduction from the DLC's value that funds the issuer's hedge against extreme overnight moves in the Underlying Asset. This is the mechanism that ensures investors cannot lose more than their initial invested capital even in the event of a severe overnight gap. Investors who close their position before the end of the Trading Day do not pay this premium.
- **Management Fee:** an ongoing fee charged by the issuer for administering the product.

The following illustration shows how these overnight fees interact with a day's return. The underlying index rises 1.00%, which would imply a gross 5x return of 5.00%:



The net DLC return is +4.96%, below the theoretical 5.00%. On a single day this difference is small. However, because fees are deducted daily and compound in the same way returns do, their drag becomes increasingly significant the longer the holding period. This reinforces the point made before, DLCs are designed for short-term trading. The combination of volatility decay and daily fee deduction makes extended holding periods structurally costly even when the underlying moves in the investor's favor.

7 Regulatory Framework (Singapore)

DLCs listed on SGX operate within a regulatory framework established by the Monetary Authority of Singapore (MAS). Understanding this framework is relevant for any investor considering trading DLCs, as it determines both the investor eligibility requirements and the obligations placed on issuers and market makers.

7.1 MAS oversight and SIP classification

DLCs are classified by MAS as Specified Investment Products (SIPs), a category reserved for financial products deemed more complex than standard equities or unit trusts. This classification reflects the leveraged, daily resetting nature of DLCs and the specific risks outlined previously. As SIPs, DLCs are subject to enhanced regulatory oversight. Issuers are required to produce and maintain a Product Highlight Sheet (PHS) and a full Base Listing Document, both of which must be made publicly available. These documents disclose the product's mechanics, fee structure, risk factors, and issuer information in a standardized format prescribed by MAS.

7.2 Customer Knowledge Assessment (CKA) / Customer Account Review (CAR)

Because DLCs are classified as SIPs, retail investors must demonstrate sufficient knowledge or experience before they are permitted to trade them. This is done through one of two routes: Customer Knowledge Assessment (CKA), a standardized assessment administered by the broker, testing the investor's understanding of leveraged products, their risks, and their mechanics. Investors who pass the CKA are approved to trade DLCs. Another possibility is the Customer Account Review (CAR), an alternative route for investors who have relevant work experience or prior trading history in SIPs or similar products. Brokers review the investor's background and may grant access without requiring the CKA.

7.3 SGX trading hours and market maker obligations

DLCs trade on SGX during standard market hours, from 9:00am to 5:00pm Singapore Time (SGT). However, since many DLC underlyings such as the HSI or S&P 500 trade on foreign exchanges in different time zones, the Designated Market Maker (DMM) will not provide bid and ask quotes when the underlying market is closed, even if SGX itself is open. The DMM is obligated under SGX rules to maintain continuous liquidity by quoting bid and ask prices during normal trading conditions. However, there are defined circumstances under which the DMM may suspend quoting:

- During the pre-market opening and for 5 minutes following the SGX open
- When the DLC is suspended from trading, including during an Airbag observation period
- When the underlying market is closed or subject to a trading halt
- When the issuer faces technical difficulties affecting its ability to hedge
- When exceptional price volatility or market disruption events occur

8 Comparison with Similar Products

DLCs share certain surface-level characteristics with other leveraged instruments, notably CFDs and Structured Warrants but differ from them in several structurally important ways. Understanding these differences helps investors identify when a DLC is the appropriate instrument for their objective, and when an alternative product may be better suited.

8.1 DLC vs CFD

Both DLCs and CFDs offer leveraged exposure to an underlying asset and settle in cash, but they differ significantly in structure, risk profile, and cost.

- **Leverage:** DLCs offer fixed leverage of 2x, 3x, 5x, or 7x applied to the daily return of the underlying. CFDs typically offer much higher leverage, commonly 10x to 20x but this is leverage applied to capital rather than to daily returns.
- **Margin and loss potential:** DLCs require no margin and losses are strictly capped at the initial invested capital. CFDs require margin deposits and critically, losses can exceed the initial investment if the position moves sharply against the investor and margin calls are not met. This makes CFDs structurally riskier for retail investors in adverse market conditions.
- **Trading venue and clearing:** DLCs are exchange-listed on SGX and cleared centrally through SGX CDP, providing price transparency and counterparty protection. CFDs are OTC, with no central clearing, the investor bears direct counterparty exposure to the broker.
- **Commission structure:** DLC brokerage is charged on the value of the position, a fraction of the total leveraged exposure. CFD commission is typically charged on the full notional contract size, making the effective transaction cost higher for the same exposure.
- **Airbag and compounding:** DLCs carry the Airbag protection mechanism and exhibit compounding. CFDs have neither, there is no intraday circuit breaker and no daily reset that creates compounding effects.

8.2 DLC vs Structured Warrants

DLCs and Structured Warrants are both exchange-listed on SGX, cleared through SGX CDP, issued by investment banks, and require no margin. However, they differ fundamentally in their pricing mechanics.

- **Pricing:** DLC prices track the underlying asset with full transparency, the price is a direct function of the underlying's daily move and the fixed leverage factor. Structured Warrant prices are influenced by multiple variables beyond the underlying price: implied volatility, time to expiry, interest rates, and dividends.
- **Time decay:** Structured Warrants are options in nature and therefore subject to time decay (theta), their value erodes as they approach expiry, even if the underlying does not move. This decay accelerates as maturity approaches. DLCs are not options and have no equivalent time decay; their only ongoing cost is the daily fee deduction.

- **Leverage fixation:** DLC leverage is always fixed at the stated factor due to the daily reset. Structured Warrant leverage is variable, it depends on the current price of the underlying relative to the strike.
- **Maturity:** DLCs have a maximum tenure of three years. Structured Warrants are typically much shorter dated, with an average maturity of three to nine months.
- **Leverage levels:** DLCs offer moderate fixed leverage of up to 7x on daily returns. Structured Warrants can offer implied leverage of approximately 5x to 50x depending on how deep in or out of the money the option is but this leverage is unstable and changes constantly with the underlying price.

8.3 DLC vs Leveraged ETFs

Leveraged ETFs are the product most structurally similar to DLCs and are worth addressing separately. Both products reset their leverage daily and are therefore subject to the same compounding and volatility decay dynamics. The key differences are:

- **Exchange and accessibility:** Leveraged ETFs are typically listed on their home exchange, US-listed leveraged ETFs on NYSE or Nasdaq, for example and may not be directly accessible to Singapore retail investors through standard SGX accounts. DLCs on SGX are directly accessible through any SGX brokerage account after passing the CKA or CAR.
- **Underlying coverage:** SGX DLCs cover a set of underlyings relevant to Asian investors, HSI, STI, HSTECH, S&P 500, Nasdaq-100, and select Singapore-listed stocks. Leveraged ETFs offer broader global coverage but typically require foreign brokerage access.
- **Fee structure:** Both products charge ongoing management fees, but DLCs additionally include the Gap Premium as an explicit overnight hedging cost, a feature that Leveraged ETFs do not have as a separate line item, though their total expense ratios may embed similar costs less transparently.
- **Regulatory classification:** Both DLCs and Leveraged ETFs are classified as SIPs under MAS, requiring CKA or CAR before trading.

9 How the Issuer Creates and Hedges DLCs

From the investor's perspective, a DLC is a straightforward leveraged exposure to an underlying asset. From the issuer's perspective, it is a structured liability that must be actively managed every single day. This section explains how Société Générale, as the primary DLC issuer on SGX, constructs the product, maintains its hedge, and manages the residual risks that the Airbag and Gap Premium are designed to address.

9.1 Constructing the DLC: the Delta Hedge

When an investor purchases a 5x Long DLC on the HSI, the issuer immediately takes on a liability: it must pay out 5× the daily return of the HSI to the investor. To remain market-neutral, to avoid being exposed to directional market risk itself the issuer simultaneously takes an offsetting position in the underlying asset. This is called a delta hedge.

In practice, the issuer does not typically buy the underlying index directly. Instead, it uses index futures or total return swaps (TRS) on the underlying, as these instruments allow rapid, cost-efficient replication of the index exposure at the required notional size. The hedge position is sized such that a 1% move in the HSI produces a gain on the hedge that exactly offsets the 5% liability owed to DLC investors.

The result is that the issuer is economically neutral to the direction of the market. Whether the HSI rises or falls, the issuer's gain or loss on its hedge position offsets its obligation to DLC holders. The issuer's primary source of revenue is not market direction, but the fees charged to investors, the Management Fee, Funding Cost, and Gap Premium described.

9.2 Daily Rebalancing of the Hedge

Because the DLC's leverage is reset to its fixed factor at the end of every Trading Day, the issuer's hedge must also be rebalanced every day at close. This is a mechanical, non-discretionary process. Consider a 5x Long DLC with an initial notional of SGD 1,000,000. The issuer holds a futures position equivalent to SGD 5,000,000 of HSI. If the HSI rises 2% during the day, the DLC notional grows to approximately SGD 1,100,000. At close, the issuer must increase its futures position to SGD 5,500,000 to maintain the 5x hedge ratio. Conversely, if the HSI falls and the DLC notional shrinks, the futures position is reduced.

This daily rebalancing has an important market dynamic consequence. The issuer is a systematic momentum trader. When markets rise, the issuer buys more futures at close. When markets fall, it sells futures at close. This rebalancing activity, aggregated across all outstanding DLCs can contribute to end-of-day price momentum in the underlying, particularly in less liquid markets. The operational cost of executing this daily rebalancing brokerage, market impact, and the bid/ask spread on the futures is passed through to investors as the Rebalancing Cost.

9.3 Managing Overnight Gap Risk

The delta hedge described above works well during trading hours, when the issuer can continuously monitor the underlying and adjust its position. However, between the close of one session and the open of the next, the issuer cannot rebalance. If the underlying moves significantly overnight, opening far below the prior close for a Long DLC, two problems arise simultaneously. First, the DLC may lose a large fraction of its value before the issuer can adjust the hedge. Second, and more critically, if the gap is severe enough, the DLC's value could theoretically fall below zero, meaning the investor's loss exceeds their initial capital. Since DLCs guarantee that investors cannot lose more than their initial investment, the issuer would be forced to absorb any excess loss.

To protect against this tail risk, the issuer purchases overnight options, typically structured as put spreads or similar derivatives that pay out in the event of a severe gap down in the underlying. These options are repriced and repurchased daily as the DLC's notional changes. The cost of this options-based overnight protection is passed directly to investors as the Gap Premium collected at the close of each Trading Day.

9.4 The Airbag as an Issuer Risk Control

The Airbag Mechanism serves an equally important function for the issuer. During a fast intraday move against the DLC's direction, the issuer's delta hedge may not keep pace with the speed of the market, particularly in illiquid conditions where futures orders cannot be filled quickly at reasonable prices. This is known as delta slippage.

By triggering the Airbag and suspending DLC trading, the issuer buys time, the 15-minute observation period to rebalance its hedge in an orderly manner at the New Observed Level, without the pressure of live DLC trading continuing against it. The suspension therefore protects both parties simultaneously: it limits the investor's intraday loss exposure while giving the issuer a controlled window to restore the hedge.

This dual function is why the Airbag is a mandatory structural feature rather than an optional risk control, it is built into the product terms precisely because the issuer requires it to manage its own hedging risk in extreme market conditions

10 DLC Trading Strategies

DLCs are precision instruments designed for short-term, directional trading. This section covers two core use cases: a tactical directional trade as the baseline application, and portfolio hedging as the more strategically sophisticated use including the mathematics of hedge ratio construction.

10.1 Tactical Directional Trade

Setup: An investor believes the HSI will rise following a US Federal Reserve rate decision. They purchase a 5x Long DLC on HSI at SGD 2.50, deploying SGD 5,000 (2,000 DLCs). Effective market exposure: $5 \times \text{SGD } 5,000 = \text{SGD } 25,000$ with no margin requirement.

Outcome if HSI rises 2%:

$$V_1 = 5,000 \times (1 + 5 \times 0.02) = \text{SGD } 5,500 \text{ (+10\%)}$$

Outcome if HSI falls 2%:

$$V_1 = 5,000 \times (1 - 5 \times 0.02) = \text{SGD } 4,500 \text{ (-10\%)}$$

Maximum loss is capped at SGD 5,000 regardless of outcome, no margin call, no additional liability. The position should be closed on the catalyst day or shortly after. Holding beyond two to three days introduces compounding drag that erodes the position even if the underlying eventually moves in the right direction, as explain in section 4.

10.2 Portfolio Hedging with a Short DLC

Objective: Temporarily reduce net equity exposure without selling the underlying position preserving the long-term holding while neutralizing short-term downside risk around a specific event or period of uncertainty.

The number of Short DLC units required to hedge a portfolio is:

$$N_{DLC} = \frac{P \times h}{L \times P_{DLC}}$$

Where:

- P = portfolio value to be hedged
- h = desired hedge ratio
- L = leverage factor of the Short DLC
- P_{DLC} = current price per DLC

Setup: An investor holds SGD 75,000 of HSI-tracking equities and wants a full hedge using a 3x Short DLC priced at SGD 3.00:

$$N_{DLC} = \frac{75,000 \times 1.0}{3 \times 3.00} = 8,333 \approx 8,300 \text{ DLCs (rounded to board lots)}$$

Capital deployed in the hedge: $8,300 \times \text{SGD } 3.00 = \text{SGD } 24,900$

	Long Portfolio	Short DLC	Net
HSI falls 4%	- SGD 3,000	+ SGD 2,988	- SGD 12
HSI rises 4%	+ SGD 3,000	- SGD 2,988	+ SGD 12

The portfolio is effectively flat in both directions. The residual SGD 12 reflects board lot rounding. The upside trade-off is explicit, the investor sacrifices participation in a rally in exchange for downside protection. This is the fundamental cost of any hedge.

Unlike a put option, which provides stable protection regardless of the path taken, a DLC hedge degrades over multiple days. Because the DLC resets its leverage daily while the portfolio does not, the ratio between the two drifts after each session. To maintain effectiveness, the investor must rebalance the hedge daily

The practical rule therefore is to use a Short DLC to hedge over days, ahead of a specific event, a data release, or a short window of elevated risk. Use a put option to hedge over weeks or months, where daily rebalancing becomes impractical and the path-dependency of the DLC works against the investor.

11 Conclusion

Daily Leverage Certificates are a sophisticated, exchange-traded leveraged product that occupy a distinctive niche in Singapore's financial market landscape. Since their introduction on SGX in July 2017, they have offered retail and institutional investors alike a transparent, margin-free way to take short-term leveraged exposure to a range of underlying assets from Asian indices such as the HSI and STI to global benchmarks like the S&P 500 and Nasdaq-100.

This document has worked through every layer of the product: the daily reset that defines its structure, the compounding effects that flow from it, the Airbag mechanism that partially protects against extreme intraday moves, the fee structure that erodes returns over time, the regulatory framework that governs access, the issuer's hedging mechanics that explain why those fees exist, and the risks that remain even after all protections are in place.

DLCs are a well-engineered product for a specific and narrow purpose: short-term, directional, event-driven trading. Used within that window, one to three days, with a defined view and a clear exit they offer a level of capital efficiency and transparency that few comparable instruments can match. Used outside it, the same daily reset mechanism that makes them powerful becomes a structural liability, compounding losses and eroding value regardless of whether the investor's long-term thesis is correct. The strategies outlined in Section 10 reflect this duality: the product's value lies not in what it does, but in knowing precisely when to use it.

12 Glossary

Airbag Mechanism: A built-in risk control feature of DLCs that suspends trading when the Underlying Asset moves adversely beyond a pre-set trigger level.

Airbag Trigger Level: The threshold adverse move in the Underlying Asset that activates the Airbag Mechanism. Varies by leverage factor and underlying type.

Base Listing Document: A regulatory document required by MAS that the issuer must publish, disclosing the terms, mechanics, fee structure, risk factors, and issuer information of the product.

Board Lot: The minimum tradeable unit of a DLC on SGX, equivalent to 100 certificates.

CDP: Central Depository, the SGX subsidiary responsible for the clearing, settlement, and custody of securities traded on SGX, including DLCs.

Customer Account Review (CAR): An alternative eligibility route for investors wishing to trade Specified Investment Products instead of requiring a CKA.

Customer Knowledge Assessment (CKA): A standardized assessment administered by brokers to verify that a retail investor has sufficient understanding of leveraged products before being permitted to trade Specified Investment Products such as DLCs.

Daily Reset: The mechanism by which a DLC's leverage exposure is restored to its fixed factor at the end of every Trading Day, using the Intrinsic Close as the new base level.

Delta Hedge: The offsetting position taken by the DLC issuer in the underlying asset, typically via futures or total return swaps, to remain market-neutral and eliminate directional exposure to the market.

Delta Slippage: The risk that the issuer's delta hedge cannot keep pace with a rapid intraday move in the underlying, creating a temporary mismatch between the hedge and the DLC liability. The Airbag Mechanism is partly designed to give the issuer time to correct this.

Designated Market Maker (DMM): The entity appointed by the DLC issuer and approved by SGX to provide bid and ask quotes for DLCs during trading hours.

Excluded Investment Product (EIP): The MAS classification for standard, lower-complexity investment products such as equities and plain vanilla bonds.

Funding Cost: An overnight fee charged to DLC holders, benchmarked to a short-term interest rate, reflecting the cost of financing the issuer's leveraged hedge position.

Gap Premium: A daily fee deducted from the DLC's value at the close of each Trading Day, used by the issuer to purchase overnight options that protect against extreme gap moves in the Underlying Asset between sessions. Not charged to investors who close their position intraday.

Intrinsic Close: The theoretical fair value of a DLC calculated by the issuer at the official close of the Underlying Asset's market, applying the leverage factor to the closing level net of daily costs. This is the correct reference price for calculating the next day's DLC performance.

Management Fee: An ongoing annual fee charged by the DLC issuer for administering the product, deducted daily from the DLC's value when held overnight.

MAS: Monetary Authority of Singapore. The central bank and financial regulator of Singapore, responsible for overseeing DLC issuers, the SIP classification, and CKA/CAR requirements.

Minimum Observed Level (MOL): The lowest level reached by the Underlying Asset during the 15-minute observation period following an Airbag trigger on a Long DLC. Equivalent to the New Observed Level for Long DLCs.

New Observed Level (NOL): The reference level recorded during the Airbag observation period, the lowest level for a Long DLC and the highest for a Short DLC. This level replaces the previous close as the base for subsequent DLC performance calculations after the Airbag triggers.

Overnight Gap: A sharp move in the Underlying Asset between the close of one session and the open of the next, occurring outside trading hours when the Airbag cannot be triggered and the issuer cannot rebalance its hedge.

Path Dependency: The property of DLC returns whereby the total return over multiple days depends not only on the overall move in the Underlying Asset but on the specific sequence of daily returns. A consequence of the daily reset and compounding mechanism.

Rebalancing Cost: An overnight fee reflecting the cost of restoring the issuer's delta hedge to the correct notional at the close of each Trading Day, passed to investors holding DLCs overnight.

SGX: Singapore Exchange. The exchange on which DLCs are listed and traded.

Total Return Swap: A derivative contract used by DLC issuers to replicate the performance of an underlying index efficiently, as an alternative to purchasing the index constituents directly.

Volatility Decay: The structural erosion of a DLC's value that occurs when the underlying asset moves in an oscillating or sideways pattern over multiple days. Also known as beta slippage. A consequence of the mathematical asymmetry of compounded gains and losses; a loss of X% requires a gain of more than X% to recover.

13 References

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